



Classification: Basic Concepts and Techniques

Based on “Introduction to Data Mining” by
Tan, Steinbach, Karpatne, Kumar

Classification: Definition

- Given a collection of records (training set)
 - Each record is by characterized by a tuple (x,y) , where x is the attribute set and y is the class label
 - ◆ x : attribute, predictor, independent variable, input
 - ◆ y : class, response, dependent variable, output
- Task:
 - Learn a model that maps each attribute set x into one of the predefined class labels y

Examples of Classification Task

Task	Attribute set, x	Class label, y
Categorizing email messages	Features extracted from email message header and content	spam or non-spam
Identifying tumor cells	Features extracted from x-rays or MRI scans	malignant or benign cells
Cataloging galaxies	Features extracted from telescope images	Elliptical, spiral, or irregular-shaped galaxies

General Approach for Building Classification Model

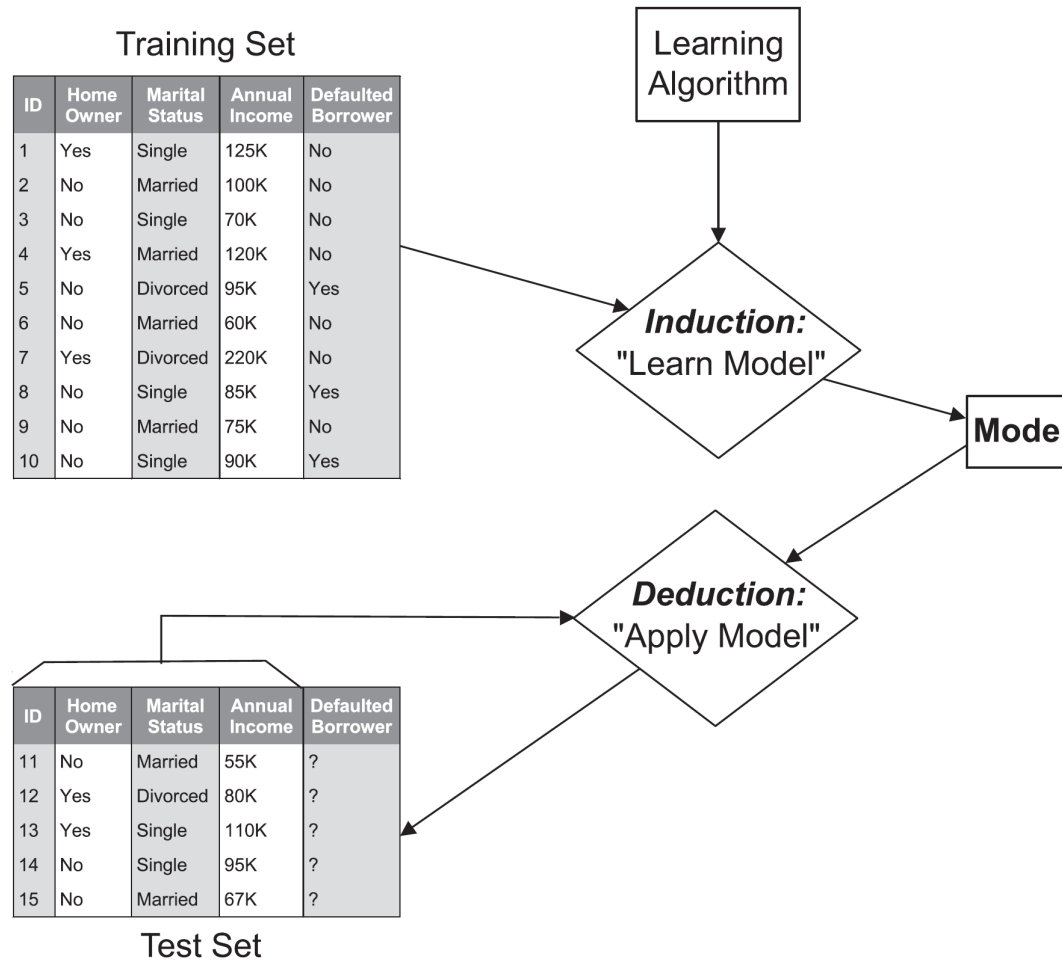


Figure 3.3. General framework for building a classification model.

Classification Techniques

- Base Classifiers
 - Decision Tree based Methods
 - Rule-based Methods
 - Nearest-neighbor
 - Naïve Bayes and Bayesian Belief Networks
 - Support Vector Machines
 - Neural Networks, Deep Neural Nets
- Ensemble Classifiers
 - Boosting, Bagging, Random Forests

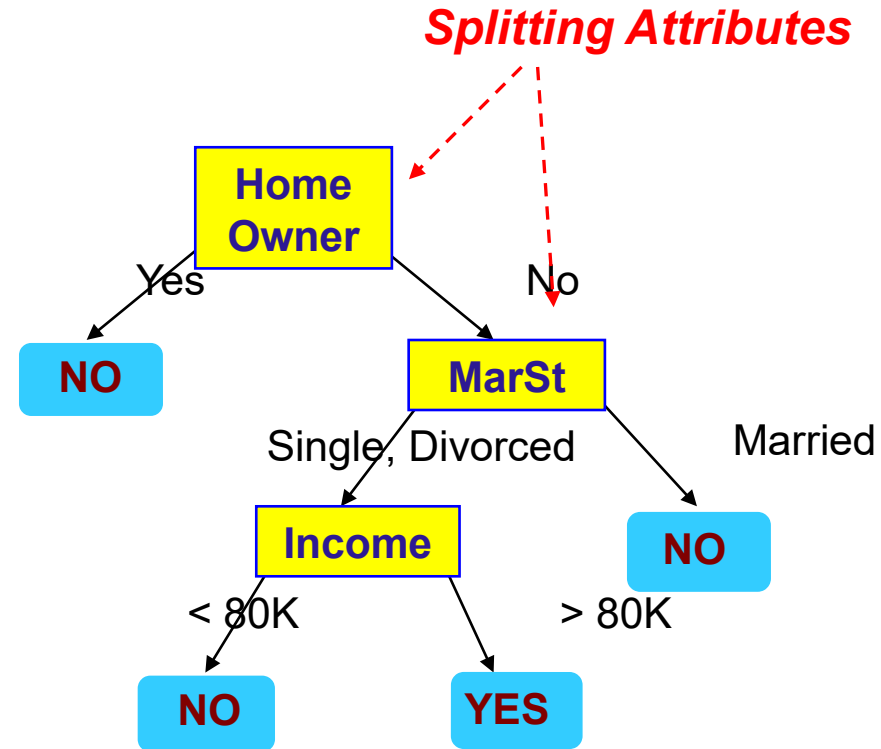
Ensemble classifiers are called "ensemble" because they combine multiple base classifiers to create a more powerful and accurate predictive model. The idea is that by aggregating the predictions from several individual classifiers, the ensemble method can reduce errors and improve performance over any single classifier.

Example of a Decision Tree

categorical
categorical
continuous
class

ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

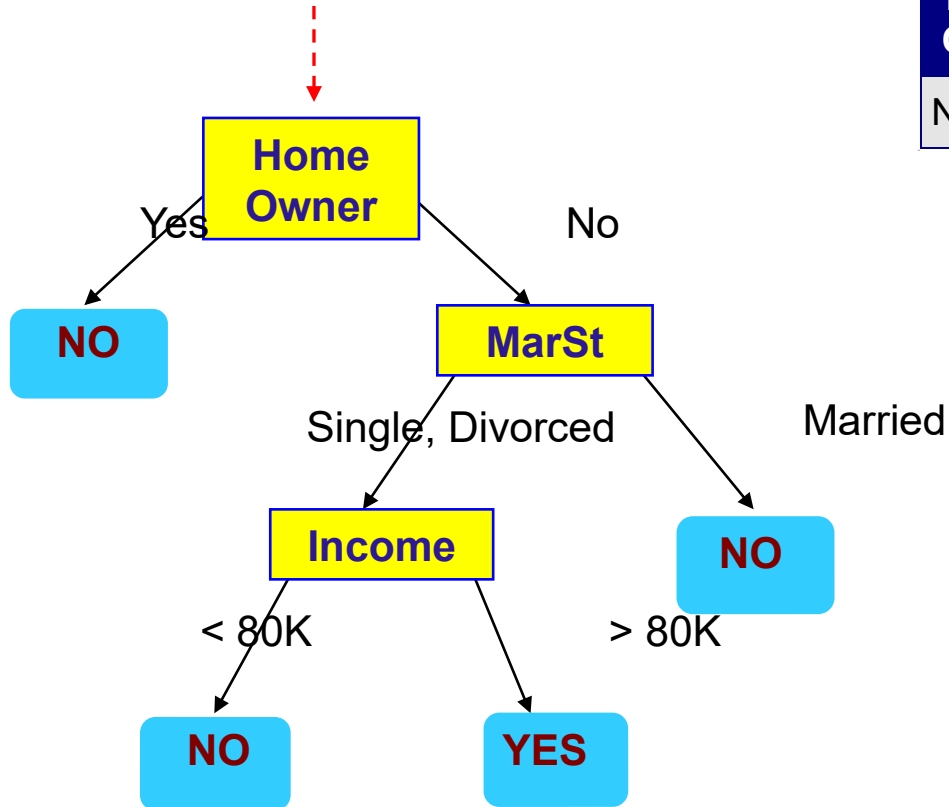
Training Data



Model: Decision Tree

Apply Model to Test Data

Start from the root of tree.



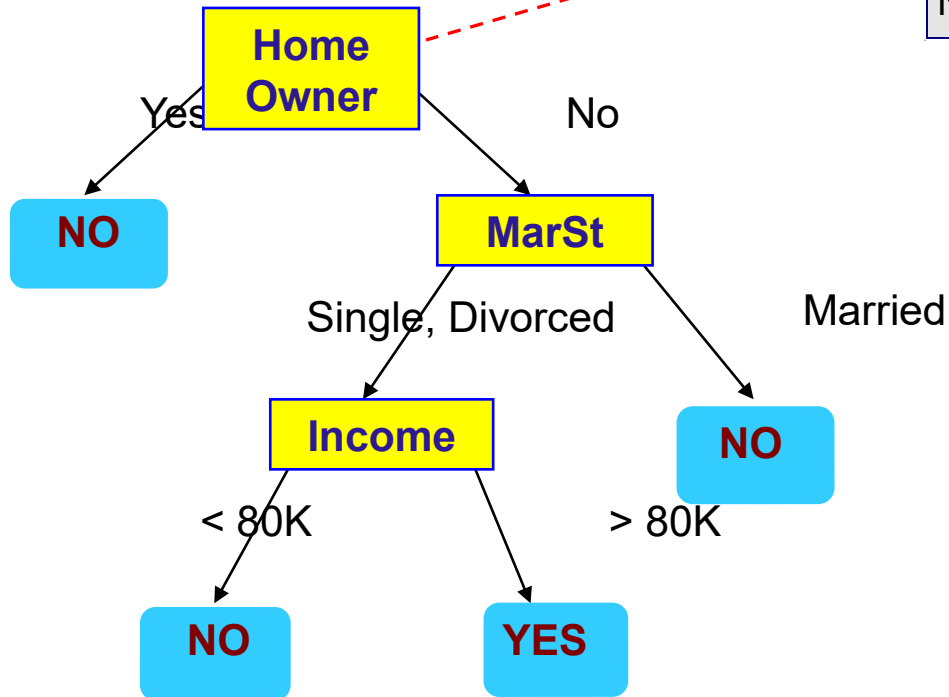
Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

Apply Model to Test Data

Test Data

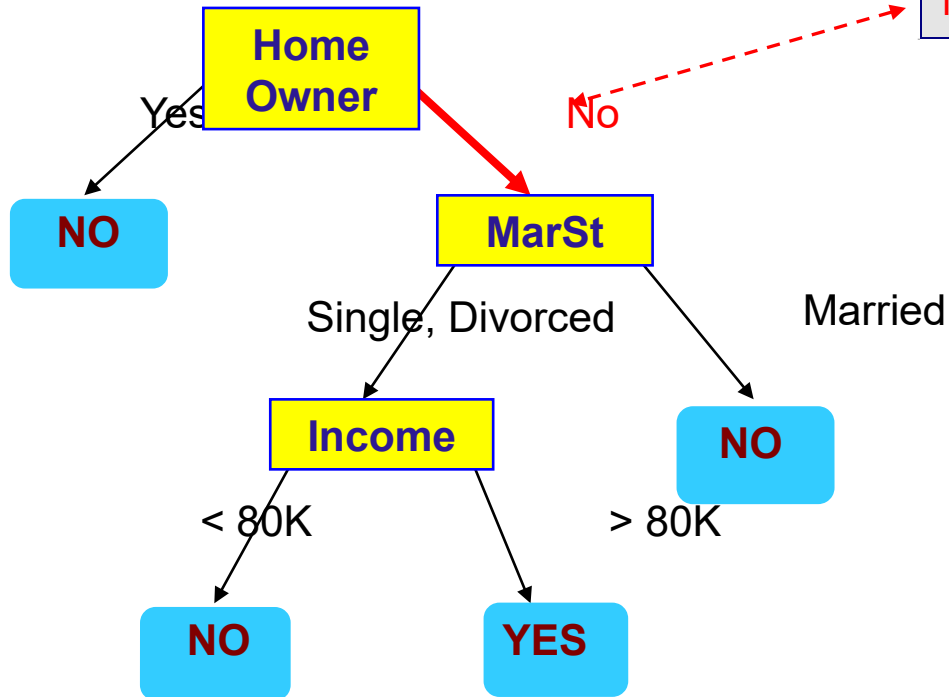
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Apply Model to Test Data

Test Data

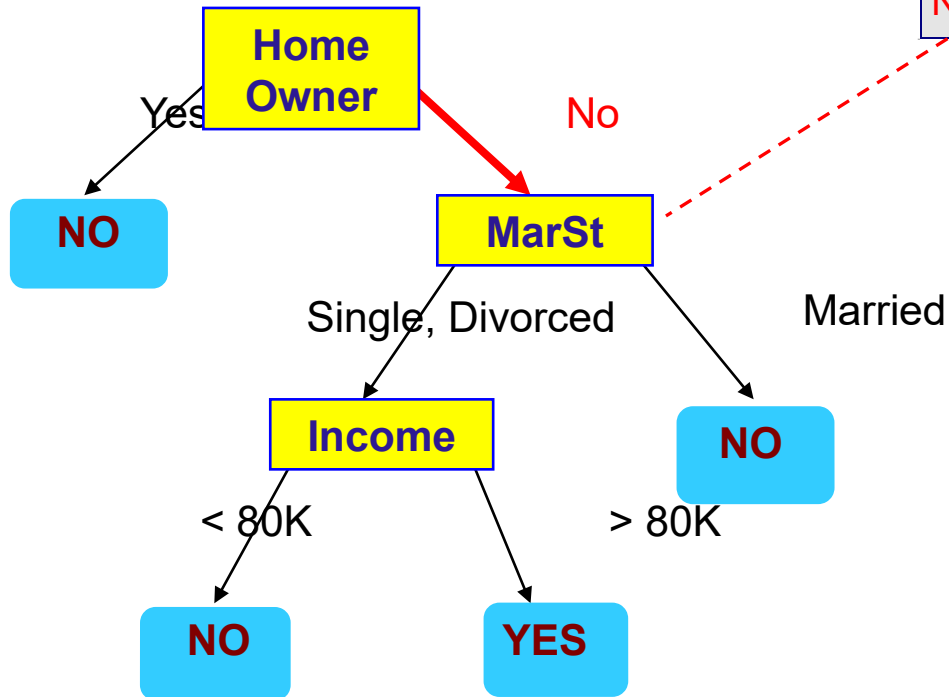
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Apply Model to Test Data

Test Data

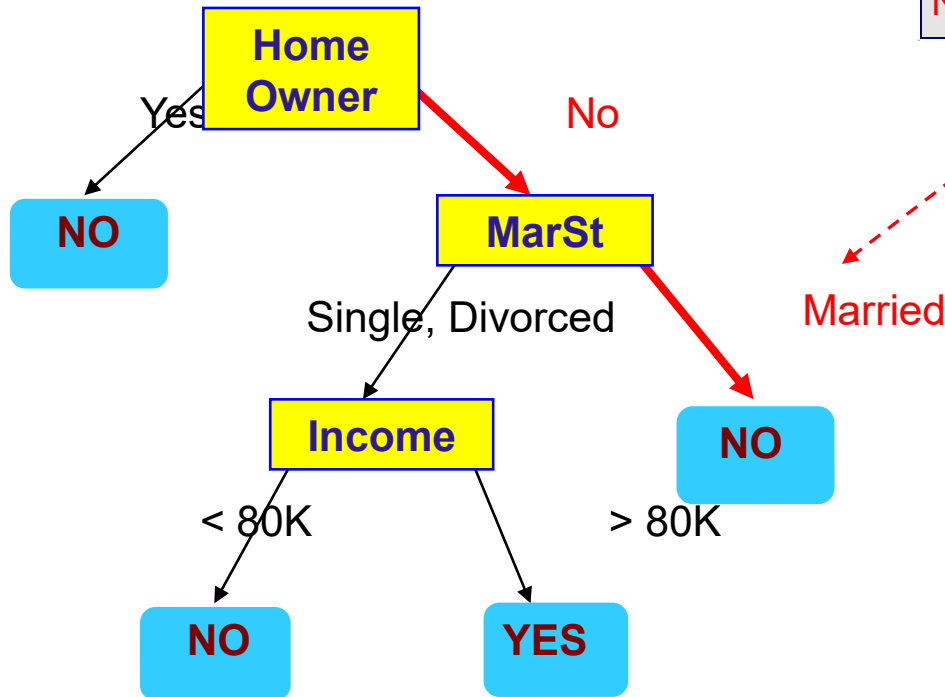
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Apply Model to Test Data

Test Data

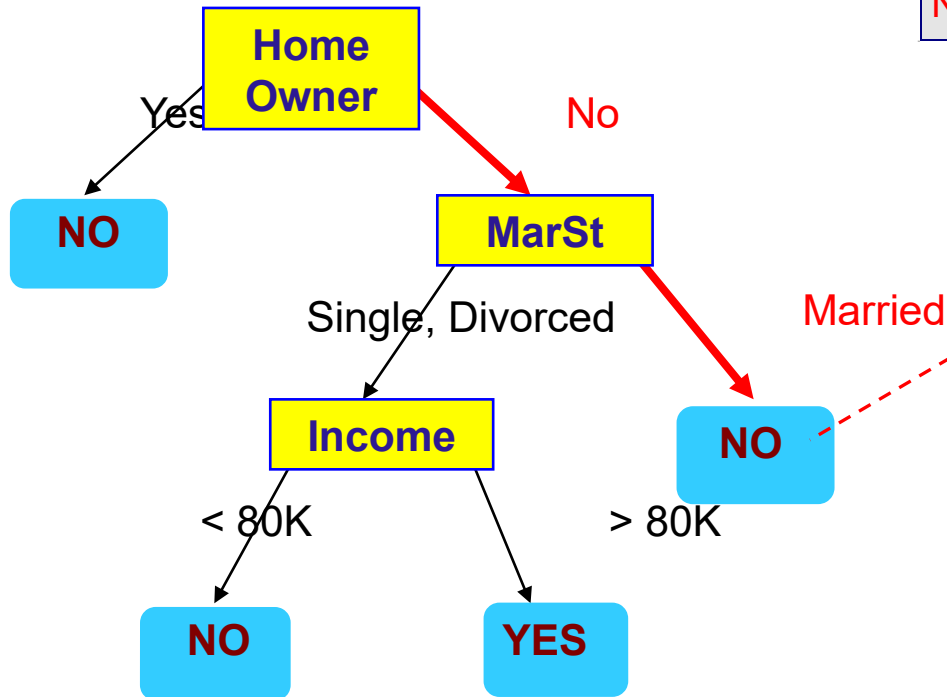
Home Owner	Marital Status	Annual Income	Defaulted Borrower
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Apply Model to Test Data

Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

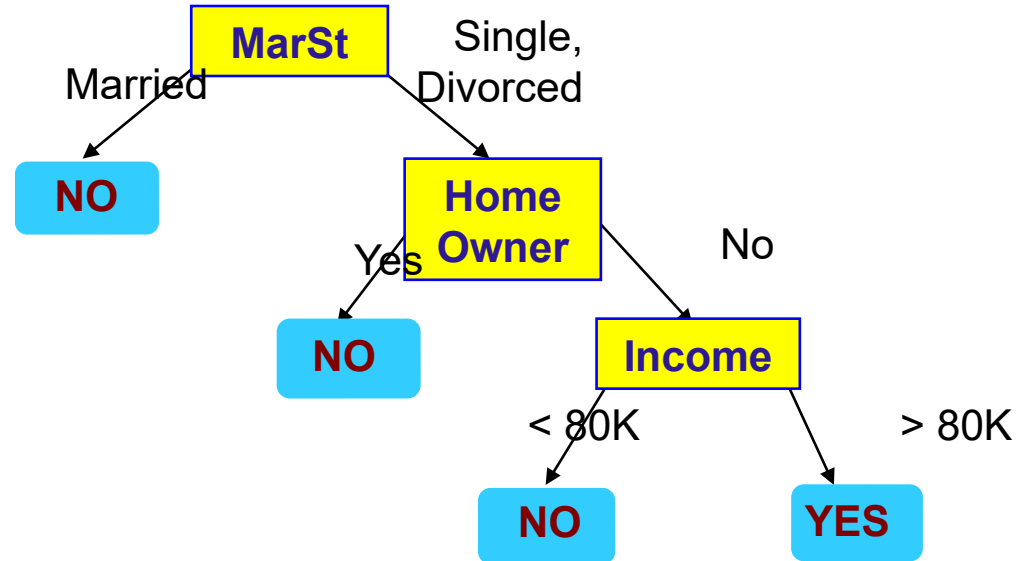


Assign Defaulted to
"No"

Another Example of Decision Tree

categorical
categorical
continuous
class

ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
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3	No	Single	70K	No
4	Yes	Married	120K	No
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6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



There could be more than one tree that fits the same data!

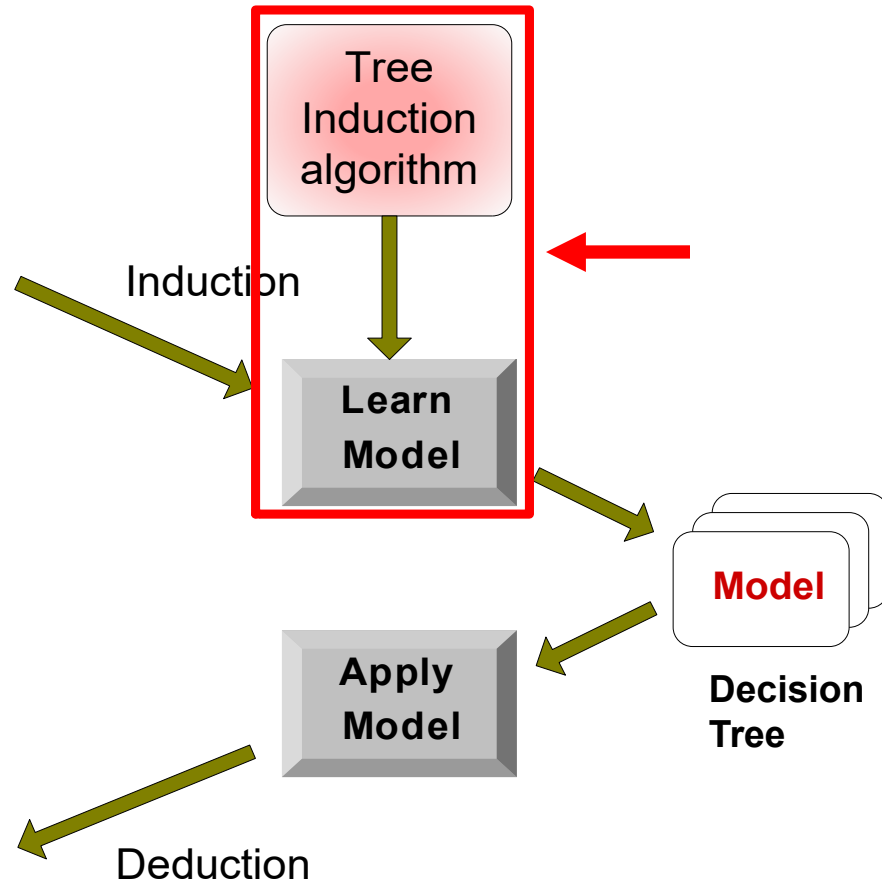
Decision Tree Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



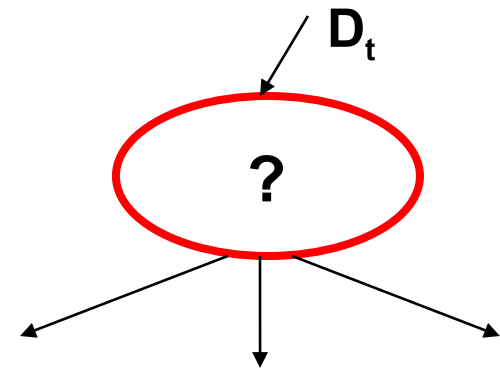
Decision Tree Induction

- Many Algorithms:
 - Hunt's Algorithm (one of the earliest)
 - CART
 - ID3, C4.5
 - SLIQ, SPRINT

General Structure of Hunt's Algorithm

- Let D_t be the set of training records that reach a node t
- General Procedure:
 - If D_t contains records that belong the same class y_t , then t is a leaf node labeled as y_t
 - If D_t contains records that belong to more than one class, use an attribute test to split the data into smaller subsets. Recursively apply the procedure to each subset.

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7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



Hunt's Algorithm

Defaulted = No

(7,3)

(a)

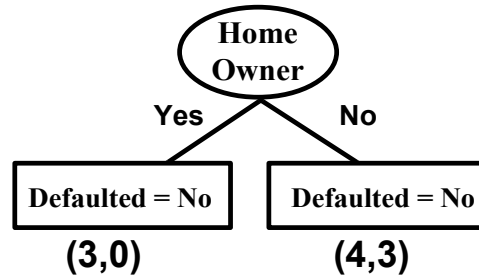
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7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Hunt's Algorithm

Defaulted = No

(7,3)

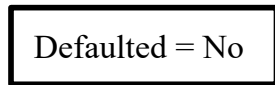
(a)



(b)

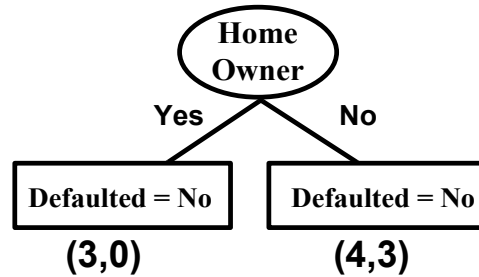
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Hunt's Algorithm



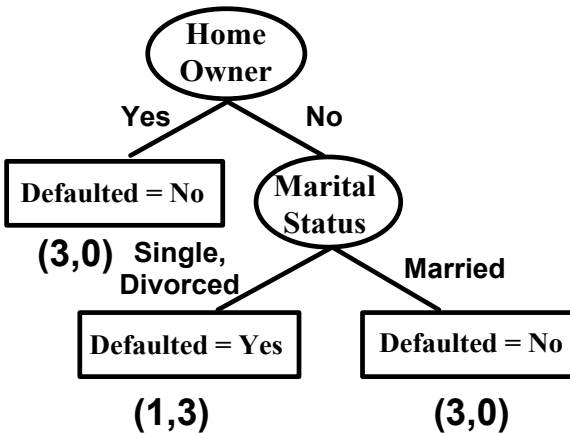
(7,3)

(a)



(b)

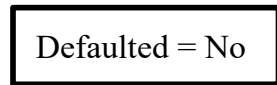
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(c)

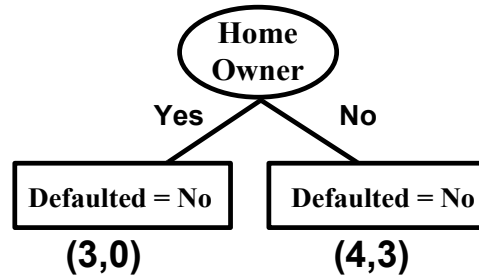
Hunt's Algorithm

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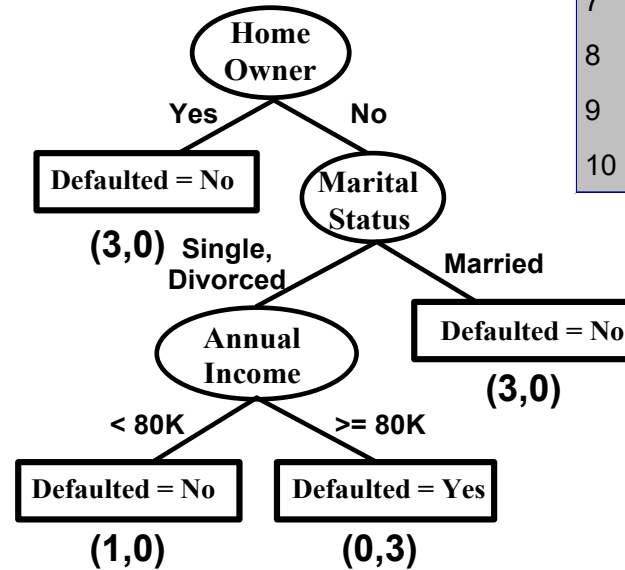


(7,3)

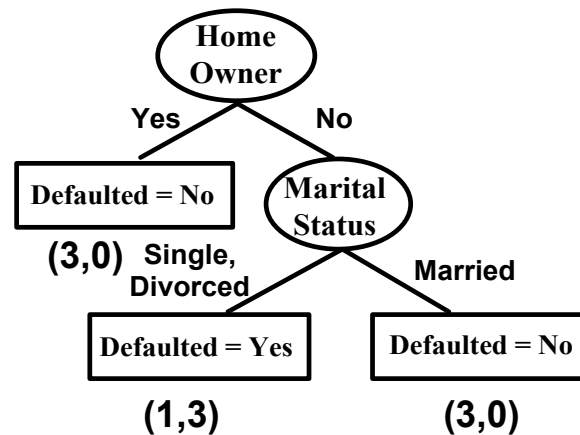
(a)



(b)



(d)



(c)

Design Issues of Decision Tree Induction

- How should training records be split?
 - Method for expressing test condition
 - ◆ depending on attribute types
 - Measure for evaluating the goodness of a test condition
- How should the splitting procedure stop?
 - Stop splitting if all the records belong to the same class or have identical attribute values
 - Early termination

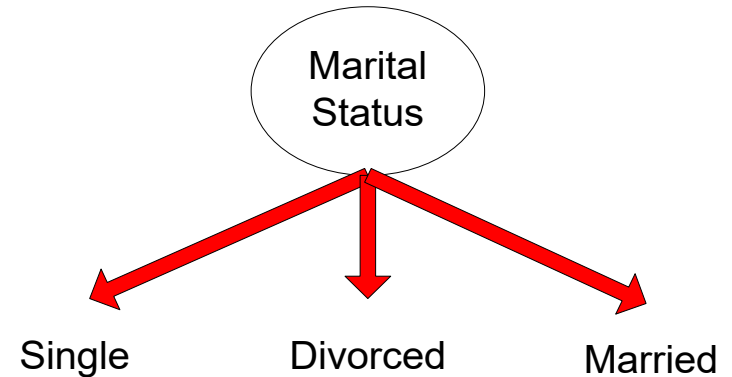
Methods for Expressing Test Conditions

- Depends on attribute types
 - Binary
 - Nominal
 - Ordinal
 - Continuous

Test Condition for Nominal Attributes

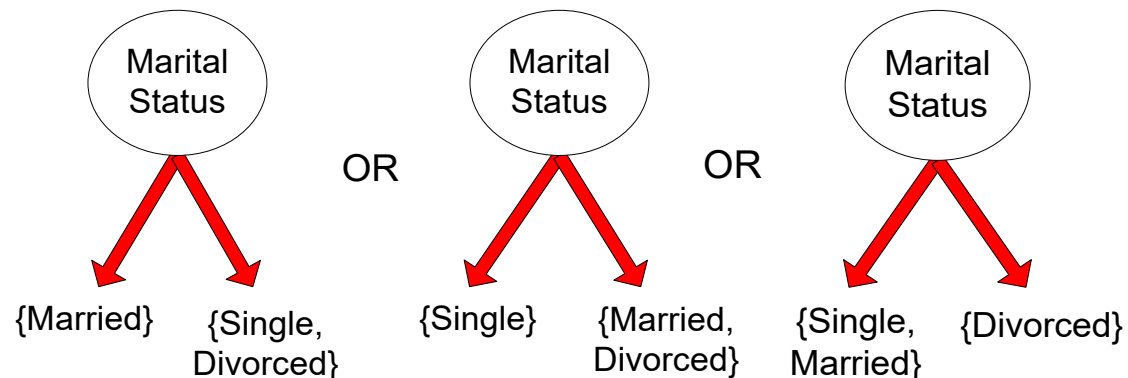
- **Multi-way split:**

- Use as many partitions as distinct values.



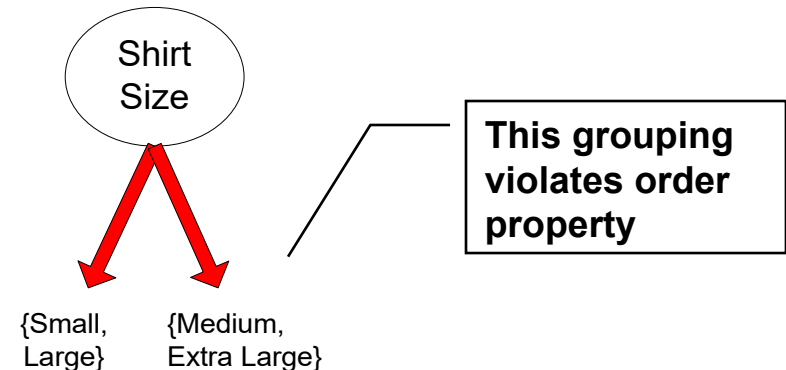
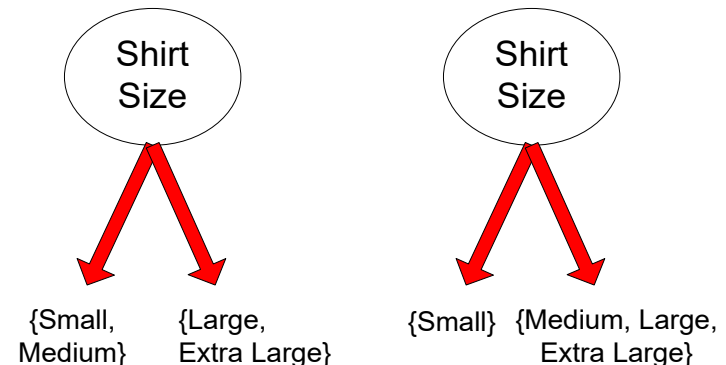
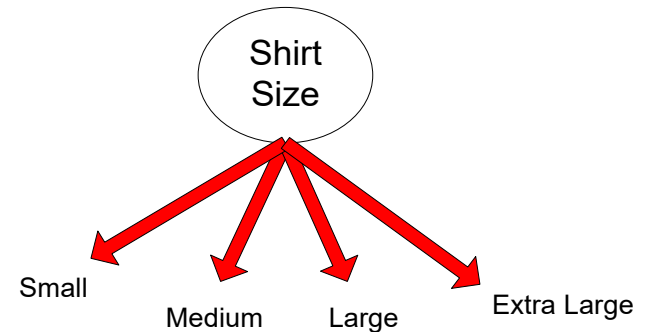
- **Binary split:**

- Divides values into two subsets

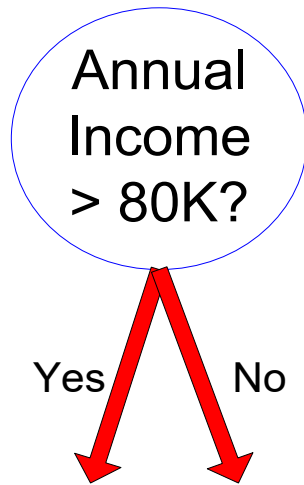


Test Condition for Ordinal Attributes

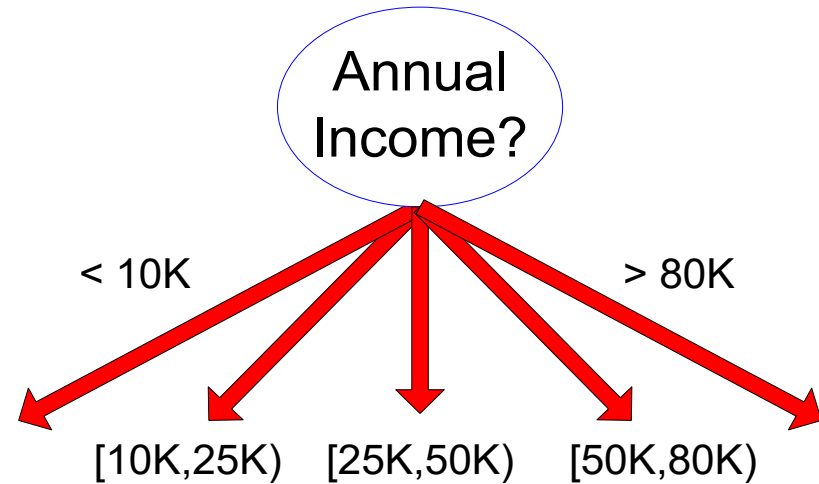
- **Multi-way split:**
 - Use as many partitions as distinct values
- **Binary split:**
 - Divides values into two subsets
 - Preserve order property among attribute values



Test Condition for Continuous Attributes



(i) Binary split



(ii) Multi-way split

Splitting Based on Continuous Attributes

- Different ways of handling
 - **Discretization** to form an ordinal categorical attribute

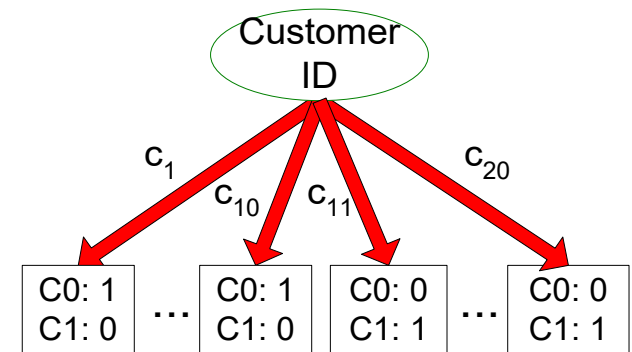
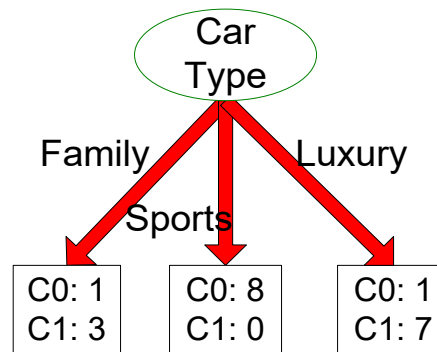
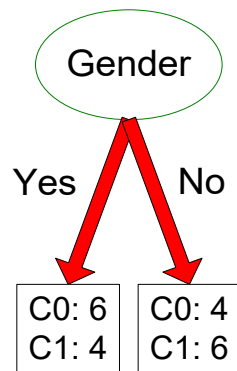
Ranges can be found by equal interval bucketing, equal frequency bucketing (percentiles), or clustering.

 - ◆ Static – discretize once at the beginning
 - ◆ Dynamic – repeat at each node
 - **Binary Decision**: $(A < v)$ or $(A \geq v)$
 - ◆ consider all possible splits and finds the best cut
 - ◆ can be more compute intensive

How to determine the Best Split

**Before Splitting: 10 records of class 0,
10 records of class 1**

Customer Id	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1



Which test condition is the best?

How to determine the Best Split

- Greedy approach:
 - Nodes with **pur**er class distribution are preferred
- Need a measure of node impurity:

C0: 5
C1: 5

High degree of impurity

C0: 9
C1: 1

Low degree of impurity

Measures of Node Impurity

- Gini Index

$$= 1 - \sum_{i=0}^{c-1} \left(\frac{f_i}{N} \right)^2$$

Where f_i is the frequency of class i at node t , and N is the total number of classes

- Entropy

$$= - \sum_{i=0}^{c-1} \left(\frac{f_i}{N} \right) \log_2 \left(\frac{f_i}{N} \right)$$

- Misclassification error

$$= 1 - \max_i \left[\frac{f_i}{N} \right]$$

Finding the Best Split

1. Compute impurity measure (P) before splitting
2. Compute impurity measure (M) after splitting
 - Compute impurity measure of each child node
 - M is the weighted impurity of child nodes
3. Choose the attribute test condition that produces the highest gain

$$\text{Gain} = P - M$$

or equivalently, lowest impurity measure after splitting (M)

Finding the Best Split

Before Splitting:

C0	N00
C1	N01

→ **P**

A?

Yes

No

Node N1

Node N2

C0

N10

C1

N11

C0

N20

C1

N21

M11

M12

M1

B?

Yes

No

Node N3

Node N4

C0

N30

C1

N31

C0

N40

C1

N41

M21

M22

M2

Gain = P – M1 vs P – M2

Measure of Impurity: GINI

- Gini Index for a given node

$$= 1 - \sum_{i=0}^{c-1} \left(\frac{f_i}{N} \right)^2$$

Where f_i is the frequency of class i at node n , and N is the total number of classes

- Maximum of 0.5 when records are equally distributed among all classes, implying the least beneficial situation for classification
- Minimum of 0 when all records belong to one class, implying the most beneficial situation for classification
- Gini index is used in decision tree algorithms such as CART, SLIQ, SPRINT

Measure of Impurity: GINI

- Gini Index for a given node t :

$$= 1 - \sum_{i=0}^{-1} \left(\quad \right)^2$$

- For 2-class problem (p, 1 – p):
 - ◆ $\text{GINI} = 1 - p^2 - (1 - p)^2 = 2p(1-p)$

C1	0
C2	6
Gini=0.000	

C1	1
C2	5
Gini=0.278	

C1	2
C2	4
Gini=0.444	

C1	3
C2	3
Gini=0.500	

Computing Gini Index of a Single Node

$$= 1 - \sum_{i=0}^{-1} \left(\quad \right)^2$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$\text{Gini} = 1 - P(C1)^2 - P(C2)^2 = 1 - 0 - 1 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$\text{Gini} = 1 - (1/6)^2 - (5/6)^2 = 0.278$$

C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

$$\text{Gini} = 1 - (2/6)^2 - (4/6)^2 = 0.444$$

Computing Gini Index for a Collection of Nodes

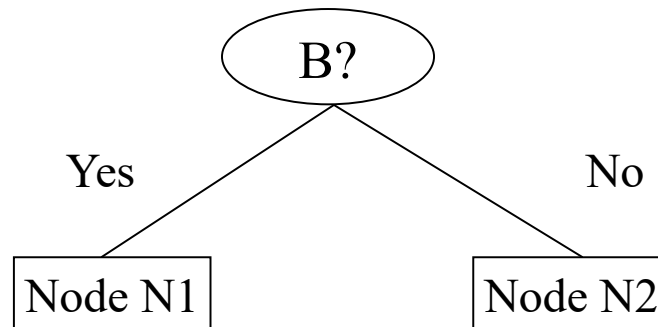
- When a node is split into partitions (children)

$$= \sum_{i=1}^n \frac{n_i}{n} \left(1 - \frac{n_i}{n} \right)$$

where, n_i = number of records at child i ,
 n = number of records at parent node .

Binary Attributes: Computing GINI Index

- Splits into two partitions (child nodes)
- Effect of Weighing partitions:
 - Larger and purer partitions are sought



	Parent
C1	7
C2	5
Gini = 0.486	

Gini(N1)

$$= 1 - (5/6)^2 - (1/6)^2$$
$$= 0.278$$

Gini(N2)

$$= 1 - (2/6)^2 - (4/6)^2$$
$$= 0.444$$

	N1	N2
C1	5	2
C2	1	4
Gini=0.361		

Weighted Gini of N1 N2

$$= 6/12 * 0.278 +$$
$$6/12 * 0.444$$
$$= 0.361$$

$$\text{Gain} = 0.486 - 0.361 = 0.125$$

Categorical Attributes: Computing Gini Index

- For each distinct value, gather counts for each class in the dataset
- Use the count matrix to make decisions

Multi-way split

	CarType		
	Family	Sports	Luxury
C1	1	8	1
C2	3	0	7
Gini	0.163		

Two-way split
(find best partition of values)

	CarType	
	{Sports, Luxury}	{Family}
C1	9	1
C2	7	3
Gini	0.468	

	CarType	
	{Sports}	{Family, Luxury}
C1	8	2
C2	0	10
Gini	0.167	

Which of these is the best?

Measure of Impurity: Entropy

- Entropy at a given node

$$= - \sum_{i=0}^{c-1} \left(\frac{f_i}{n} \right) \log_2 \left(\frac{f_i}{n} \right)$$

Where f_i is the frequency of class i at node n , and c is the total number of classes

- ◆ Maximum of $\log_2 c$ when records are equally distributed among all classes, implying the least beneficial situation for classification
- ◆ Minimum of 0 when all records belong to one class, implying most beneficial situation for classification
- Entropy based computations are quite similar to the GINI index computations

Computing Entropy of a Single Node

$$= - \sum_{i=0}^{-1} \left(\begin{array}{c} \\ \end{array} \right)_2 \left(\begin{array}{c} \\ \end{array} \right)$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$\text{Entropy} = - 0 \log 0 - 1 \log 1 = - 0 - 0 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$\text{Entropy} = - (1/6) \log_2 (1/6) - (5/6) \log_2 (5/6) = 0.65$$

C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

$$\text{Entropy} = - (2/6) \log_2 (2/6) - (4/6) \log_2 (4/6) = 0.92$$

Computing Information Gain After Splitting

- Information Gain:

$$= H(S) - \sum_{i=1}^n \frac{|S_i|}{|S|} H(S_i)$$

Parent Node, is split into partitions (children)

is number of records in child node

- Choose the split that achieves most reduction (maximizes GAIN)
- Used in the ID3 and C4.5 decision tree algorithms
- Information gain is the mutual information between the class variable and the splitting variable

Measure of Impurity: Classification Error

- Classification error at a node

$$(\text{ }) = 1 - \max [(\text{ })]$$

- Maximum of when records are equally distributed among all classes, implying the least interesting situation
- Minimum of 0 when all records belong to one class, implying the most interesting situation

Computing Error of a Single Node

$$\left(\quad \right) = 1 - \max \left[\left(\quad \right) \right]$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$\text{Error} = 1 - \max(0, 1) = 1 - 1 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$\text{Error} = 1 - \max(1/6, 5/6) = 1 - 5/6 = 1/6$$

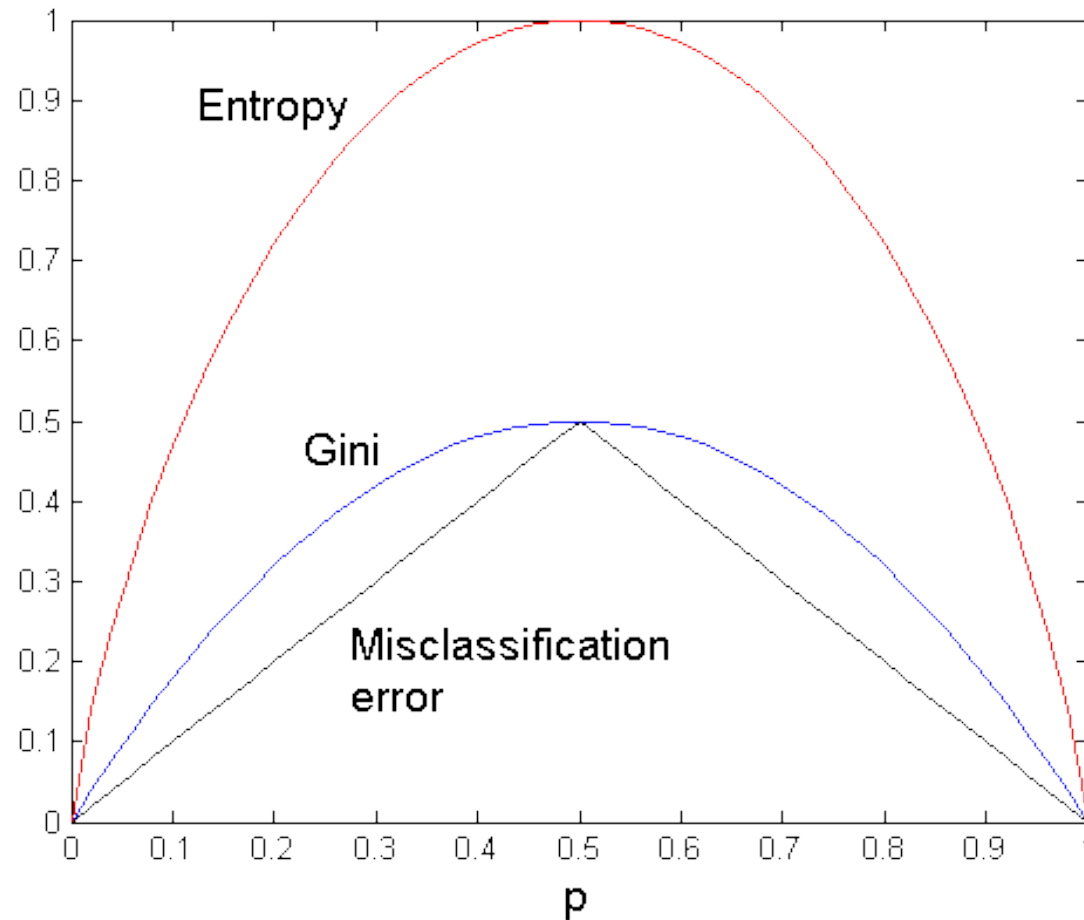
C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

$$\text{Error} = 1 - \max(2/6, 4/6) = 1 - 4/6 = 1/3$$

Comparison among Impurity Measures

For a 2-class problem:



Decision Tree Based Classification

- Advantages:

- Relatively inexpensive to construct
- Extremely fast at classifying unknown records
- Easy to interpret for small-sized trees
- Robust to noise (especially when methods to avoid overfitting are employed)
- Can easily handle redundant attributes
- Can easily handle irrelevant attributes (unless the attributes are **interacting**)

- Disadvantages: .

- Due to the greedy nature of splitting criterion, **interacting** attributes (that can distinguish between classes together but not individually) may be passed over in favor of other attributed that are less discriminating.
- Each decision boundary involves only a single attribute